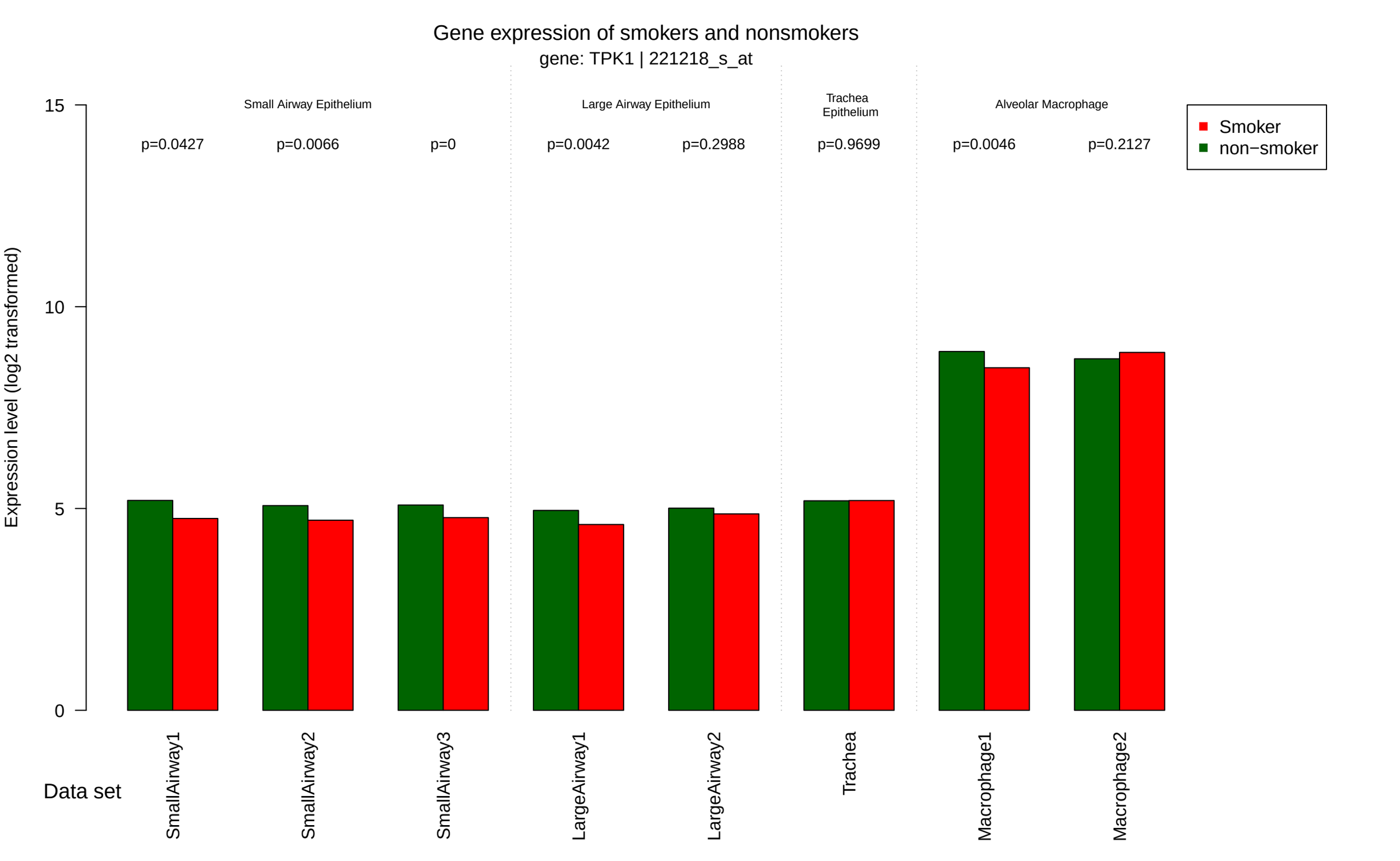
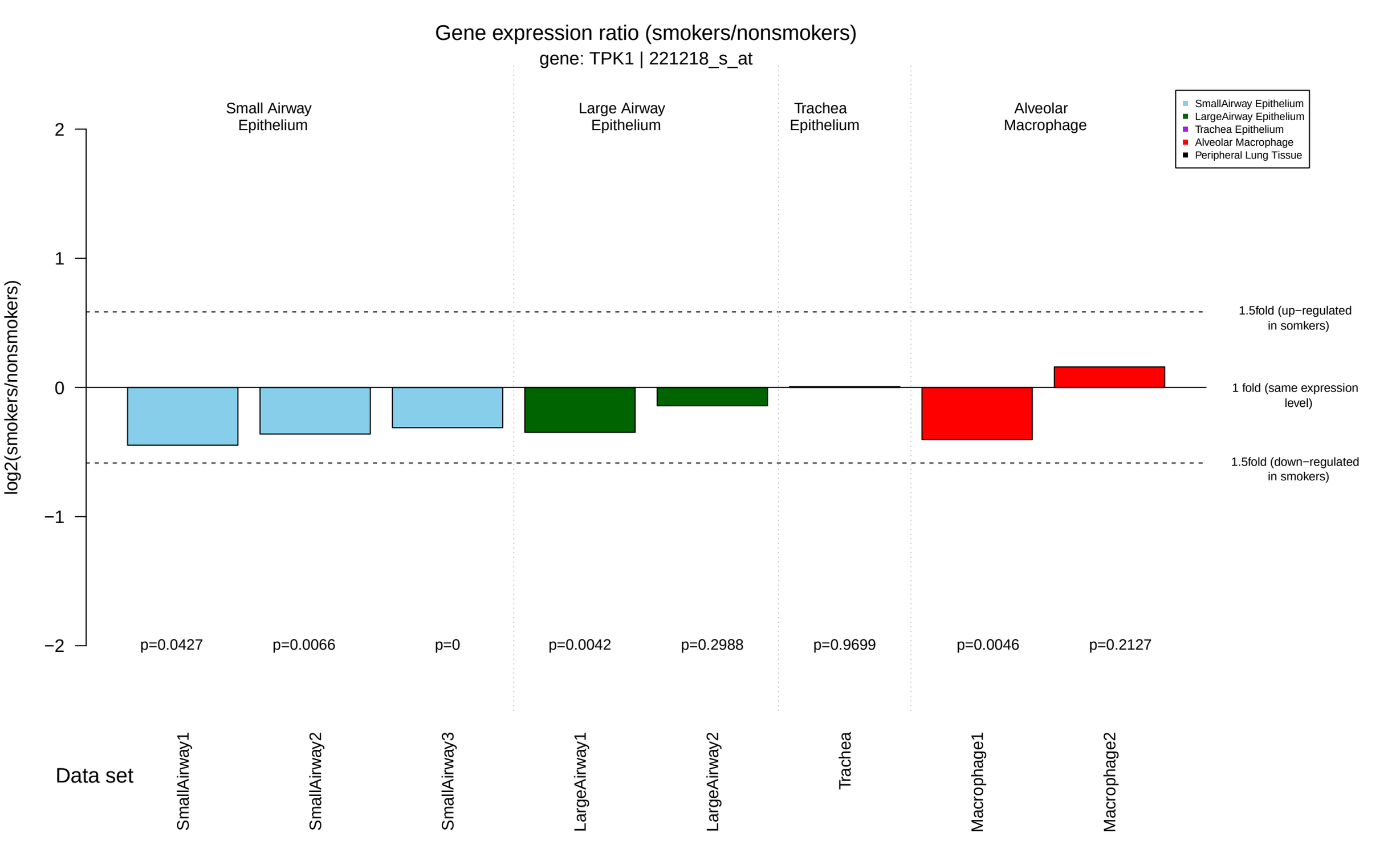
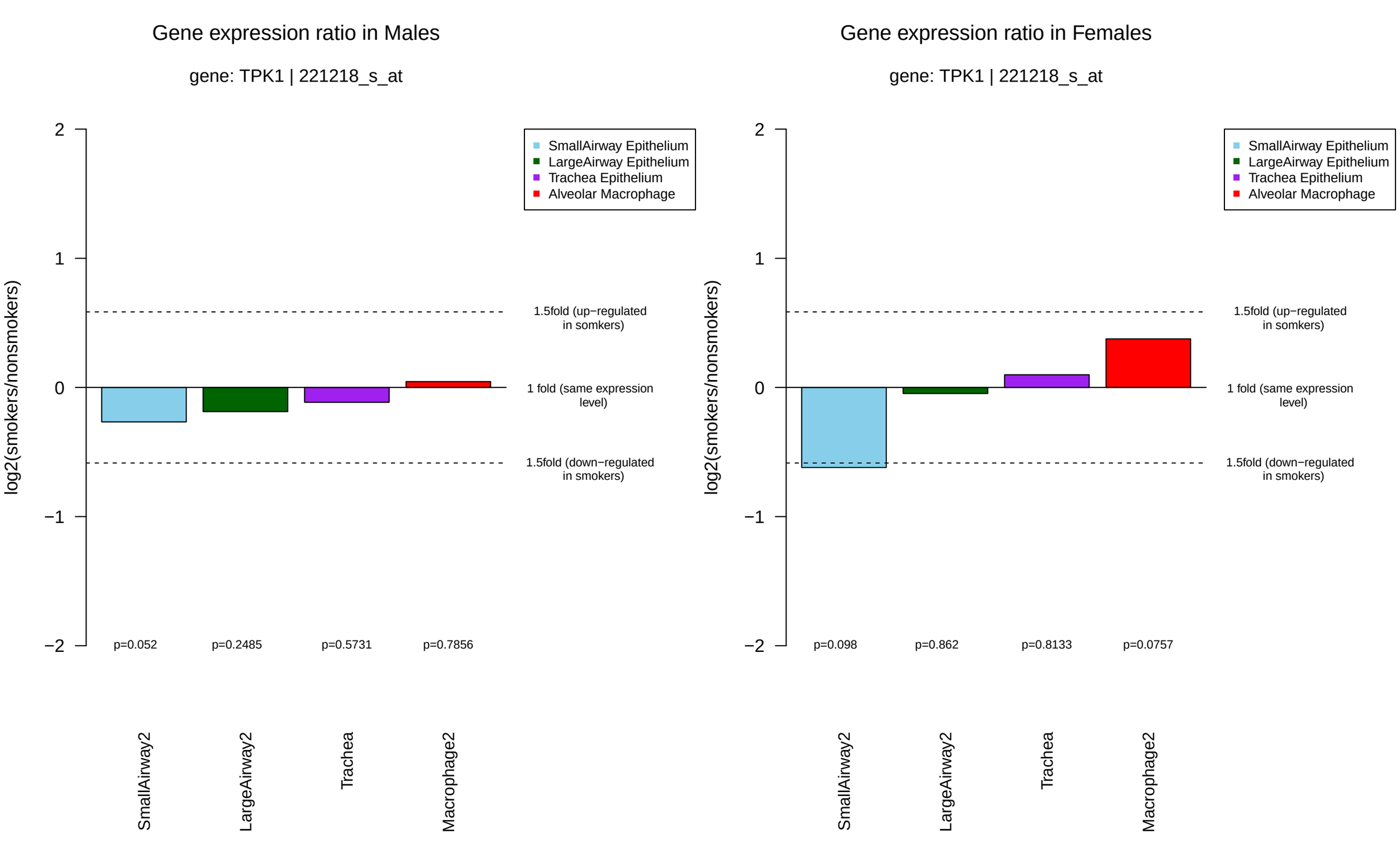
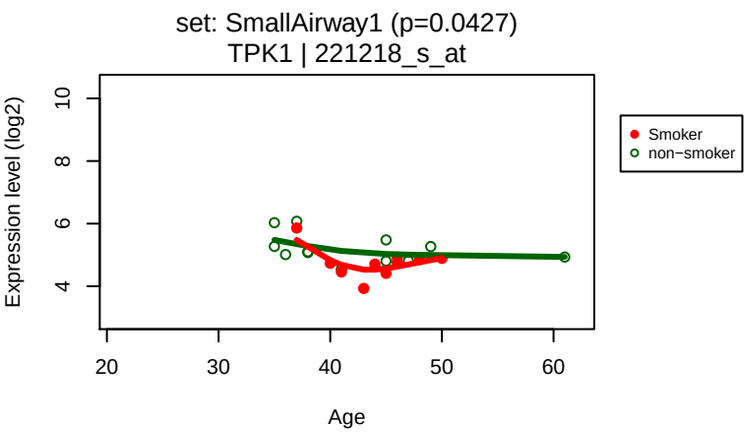
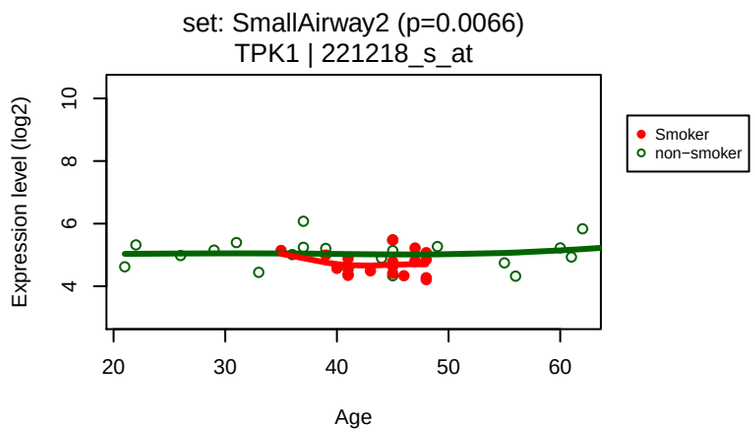
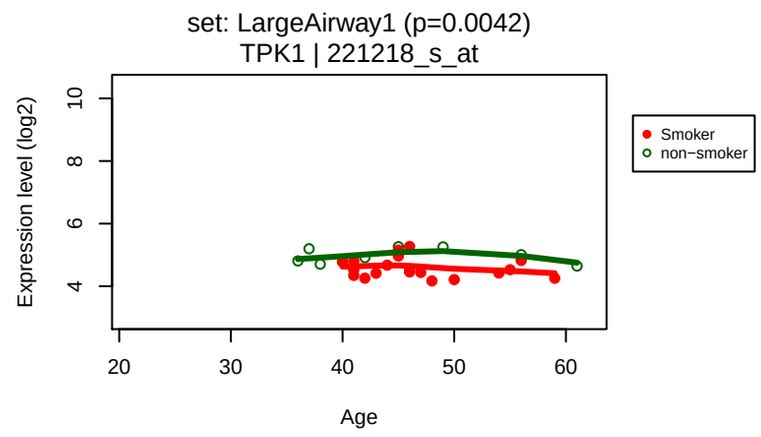

Gene expression of smokers and nonsmokers
gene: TPK1 | 221218_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0427
p=0.0066
p=0
p=0.0042
p=0.2988
p=0.9699
p=0.0046
p=0.2127
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

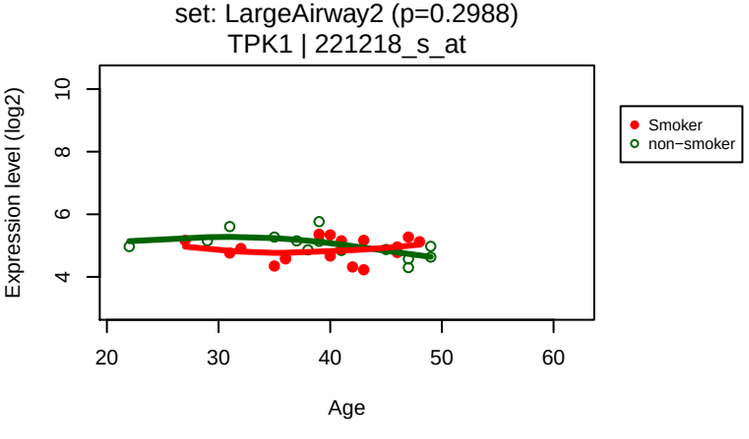

Gene expression ratio (smokers/nonsmokers)
gene: TPK1 | 221218_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0427
p=0.0066
p=0
p=0.0042
p=0.2988
p=0.9699
p=0.0046
p=0.2127
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

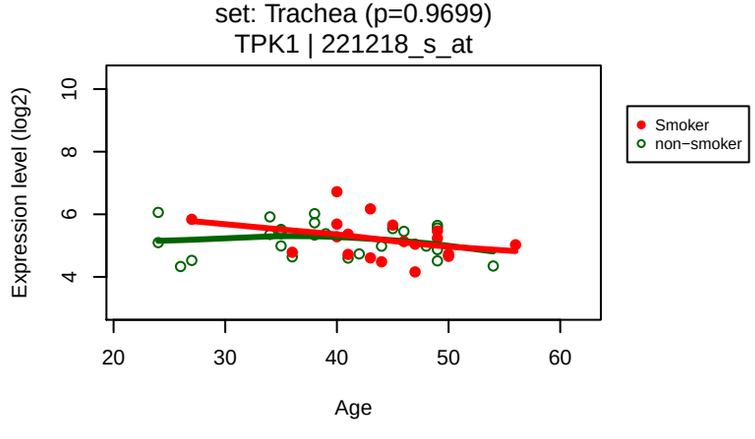

Gene expression ratio in Males
gene: TPK1 | 221218_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.052
p=0.2485
p=0.5731
p=0.7856
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TPK1 | 221218_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.098
p=0.862
p=0.8133
p=0.0757
SmallAirway2
LargeAirway2
Trachea
Macrophage2

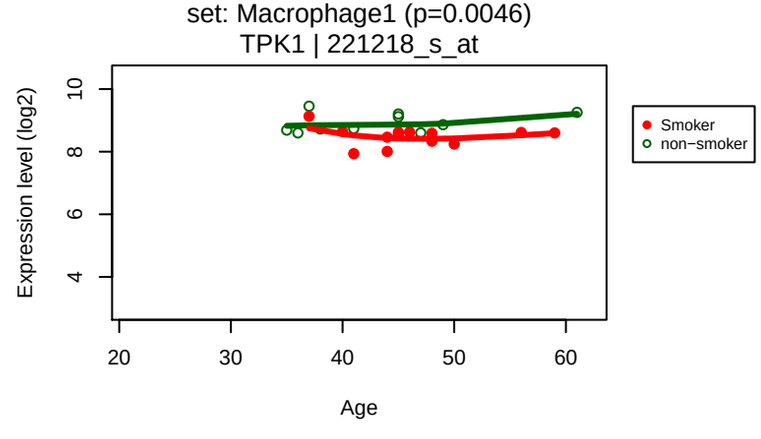

set: SmallAirway1 (p=0.0427)
TPK1 | 221218_s_at
Expression level (log2)
Age
Smoker
non−smoker

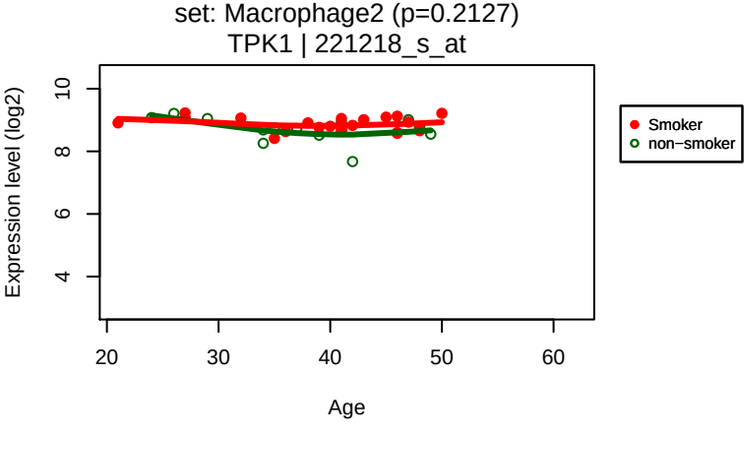

set: SmallAirway2 (p=0.0066)
TPK1 | 221218_s_at
Expression level (log2)
Age
Smoker
non−smoker

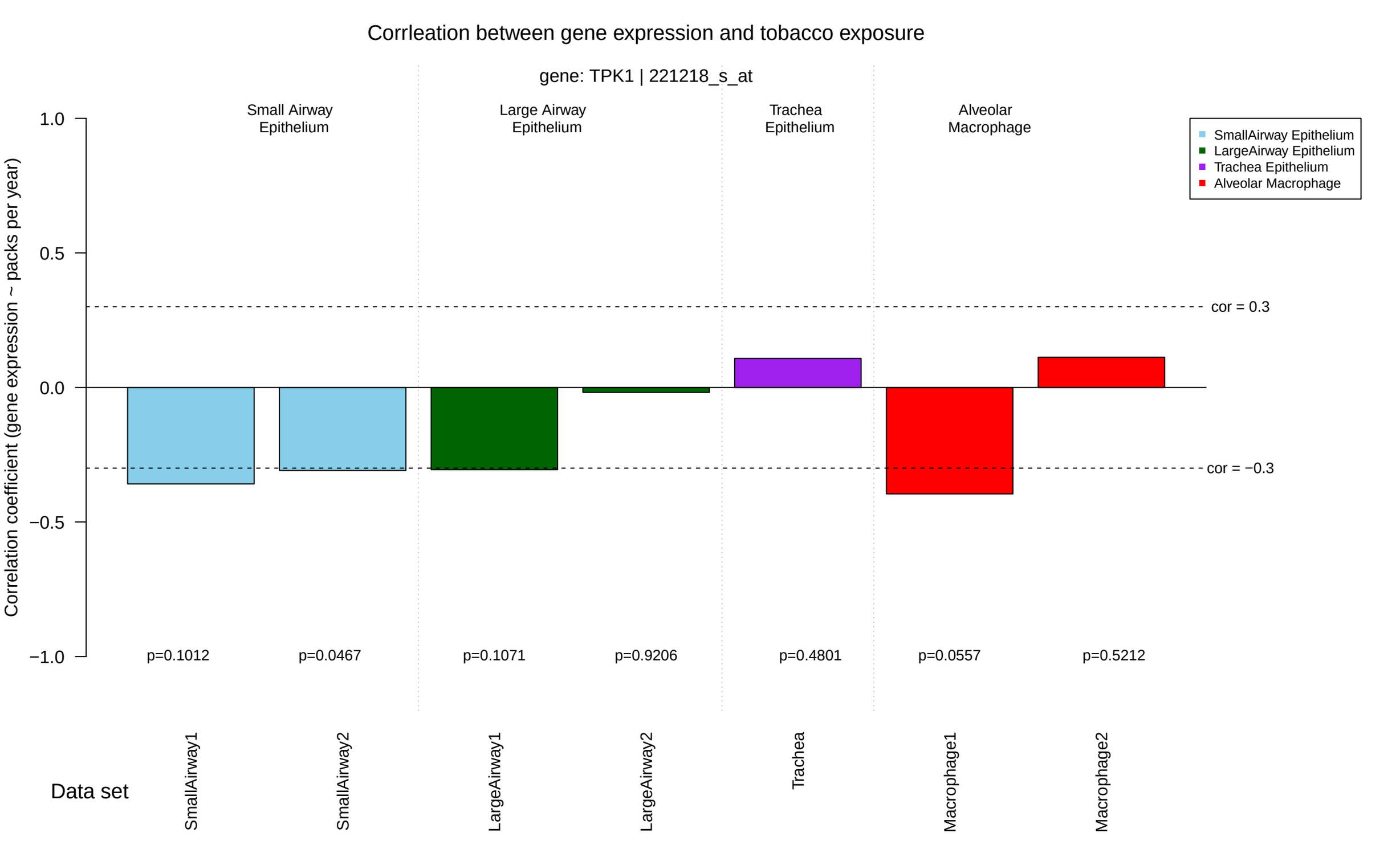

set: LargeAirway1 (p=0.0042)
TPK1 | 221218_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2988)
TPK1 | 221218_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9699)
TPK1 | 221218_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0046)
TPK1 | 221218_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2127)
TPK1 | 221218_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TPK1 | 221218_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1012
p=0.0467
p=0.1071
p=0.9206
p=0.4801
p=0.0557
p=0.5212
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2